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An empirical analysis of data-driven supply chain optimization in retail and logistics

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ABSTRACT

This study investigates the integration of big data and artificial intelligence (AI) to optimize decision-making in interconnected retail and logistics operations. While previous research often treats these domains in isolation, this paper proposes a unified framework for sales forecasting, delivery risk classification, and customer segmentation. Using the DataCo SMART Supply Chain dataset over 180,000 records (n = 180,519), we apply machine learning techniques including Random Forest and XGBoost. To ensure methodological rigor, we address common pitfalls in supply chain analytics by removing data-leakage variables to establish realistic performance baselines. The revised models achieve robust predictive performance for delivery delays (AUC = 0.97) and profit estimation (R^2 = 0.99), correcting the inflated metrics observed in unconstrained models. Furthermore, we employ K-Means clustering to identify four distinct customer segments for targeted strategy development. Finally, we propose a theoretical extension of the Technology Acceptance Model (TAM) to account for supply chain complexity, offering avenues for future empirical validation. This research contributes a reproducible workflow for vetting AI models before deployment in live supply chain environments.

1. Introduction

The massive digitalization of global industries has introduced a new world of data accessibility [1], radically changing the way business's function [2,3]. In the retail and logistics sectors, the convergence of big data and Artificial Intelligence (AI) offers unprecedented opportunities to streamline routing, optimize inventory, and predict consumer demand [4]. However, significant gaps remain in retail analytics (focused on promotions and segmentation) [5] and logistics analytics (focused on delivery efficiency) [6], which are often studied in silos.

This study aims to bridge that gap. We utilize the DataCo SMART Supply Chain dataset to simulate an end-to-end analytics framework. Unlike prior studies that rely on isolated datasets [7], this research integrates customer profiling with logistics performance to provide a holistic view of the value chain. Crucially, the paper addresses a pervasive issue in applied AI: the risk of overfitting and target leakage in synthetic datasets [8]. Reviewers and practitioners often cite "perfect" accuracy as a red flag for model validity [9,10]. Consequently, this study shifts focus from claiming "perfect" prediction to demonstrating a robust, reproducible methodology for feature selection and model validation. We also introduce a conceptual extension to the Technology Acceptance Model (TAM) [11], proposing that "Data Quality" and "Operational Complexity" are critical antecedents to AI adoption in this sector.

This study is limited by the analysis of the DataCo SMART Supply chain database, which contains over 180,000 records of orders, inventory, customer profiles, and logistics performance indicators. While the DataCo dataset is synthetic, this characteristic is leveraged as a strength rather than a limitation. Real-world supply chain data is often plagued by "noise," missing logs, and manual entry errors that obscure algorithmic performance. Using a structured, synthetic dataset, we establish a controlled "clean room" environment. This allows us to determine the theoretical upper bound of algorithmic performance through a "Digital Twin" simulation before introducing the entropy of real-world operations.

1.1. Research questions

To guide this inquiry, this study addresses three core research questions:

- How can data leakage in supply chain datasets be systematically identified and mitigated to establish realistic performance baselines?

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- To what extent can integrated retail and logistics data improve the prediction of delivery risks and profit margins compared to siloed approaches?
- How do "Data Quality" and "Operational Complexity" influence the theoretical adoption of AI in supply chains, and can these be integrated into the Technology Acceptance Model (TAM)?

1.2. Methodological summary

This paper adopts an empirical perspective to examine AI's transformational capabilities. The study combines data preprocessing, predictive modeling (Random Forest and XGBoost), and unsupervised clustering. We employ a rigorous "leakage audit" to remove post-event variables, establishing a realistic baseline for risk prediction. This systematic strategy ensures a thorough examination of the relationship between retail and logistics.

1.3. Paper structure

The remainder of this paper is organized as follows: 2 synthesizes existing literature and presents a taxonomy of current research gaps. 3 details the methodology, including data preprocessing and algorithm configuration. 4 presents the empirical results, including leakage audits and predictive metrics. 5 discusses the findings, theoretical implications for the TAM framework, and managerial strategies. Finally, 6 concludes the study with limitations and future directions.

2. Literature review

2.1. Evolution of big data in retail and logistics

The use of big data in retail and logistics represents a paradigm shift in traditional data management, transitioning to sophisticated data-enabled analytics that have transformed operational strategies [12]. Big data has transformed retail through dynamic pricing and segmentation [12]. Simultaneously, logistics has benefited from route optimization and automated warehousing [6]. However, widely cited frameworks often fail to integrate these. For instance, while McKinsey highlights inventory optimization [13,14], Rainy focuses on marketing analytics [15], and a few empirical studies examine how customer segmentation decisions affect downstream delivery risks [16].

2.2. Challenges in supply chain AI

The application of Machine Learning (ML) in Supply Chain Management (SCM) is often hampered by data quality and validation issues [17]. Synthetic datasets, while useful for simulation, lack the noise of real-world environments, leading to inflated performance metrics [8]. This study explicitly acknowledges these limitations, positioning the analysis as a "Digital Twin" simulation rather than a field experiment, filling the empirical gap identified by Zhao et al. [7,18].

2.3. Theoretical framework: augmented TAM

The proposed study seeks to address this gap by developing an augmented Technology Acceptance Model (TAM) of AI in the supply chain context, grounded in Davis's (1989) [11] original model. Foundational work by Venkatesh and Davis (2000) [19] established that "Output Quality" is a key determinant of Perceived Usefulness. Building on this, we propose that in high-stakes logistics, Data Quality acts as a "gatekeeper" construct. If the data is perceived as prone to leakage or error, the perceived usefulness collapses regardless of the algorithm's sophistication. This modification enhances the model's applicability to supply chain environments.

2.4. Taxonomy of relevant literature

Table 1 compares this study against relevant alternatives in the literature to highlight the research gap. This study will fill the gaps, thereby contributing to both industry and academia by providing a comprehensive view of supply chain optimization [15,20,21].

3. Methodology

This paper adopts an empirical perspective to examine the transformational capabilities of big data and artificial intelligence (AI) in optimizing retail and logistics activities. This paper adopts an empirical perspective to examine the transformational capabilities of big data and artificial intelligence (AI) in optimizing retail and logistics activities. The study uses a DataCo SMART Supply Chain to present the research's primary objectives, including sales forecasting, optimal customer segmentation, and logistics delivery optimization via advanced machine learning. The study approach combines data preprocessing, model prediction, clustering, and regression analysis to provide a comprehensive framework that yields actionable information. Such a systematic strategy ensures a thorough examination of the relationship between retail and logistics, providing empirical evidence that can support AI-based decision-making in supply chain management.

3.1. Dataset description, cleaning, and preprocessing

The study uses the DataCo SMART Supply Chain dataset (From Kaggle), comprising over 180,000 records that simulate real-life retail and logistics processes. To ensure data integrity, we performed the following preprocessing steps:

- ➔ Irrelevant Features: Identifiers such as Customer Email, Password, and Product Image were removed.
- ➔ Leakage Removal: Following a rigorous feature audit, variables that inherently reveal the target outcome were dropped. Specifically, Delivery Status (which explicitly indicates whether a delivery was late) and Days for shipping (actual) were removed from the predictor set to prevent target leakage.
- ➔ Handling Missing Values: Incomplete records were dropped to maintain data integrity.
- ➔ Feature engineering: Categorical data, such as Shipping Mode and Order Region, were converted into numerical values using LabelEncoder. StandardScaler was used to scale continuous variables and ensure consistent scaling across algorithms that model feature magnitudes, such as Random Forest and XGBoost [22].

3.2. Machine learning configuration

We employed a dual-approach strategy utilizing ensemble methods known for robustness in supply chain tasks [23]:

Table 1
Taxonomy of Literature Review.

Author(s)	Focus Area	Methodology	Key Limitation
Rainy (2025)	Retail Marketing	Marketing Analytics	Lacks logistics/delivery integration.
Olayinka (2021)	Logistics Tracking	Real-time Analytics	Focuses on tracking, not profit optimization.
Zhao et al. (2023)	Resilience	Structural Modeling	Theoretical; lacks predictive implementation.
Seyedan & Mafakheri (2020)	Demand Forecasting	Big Data Review	Highlights synthetic data issues but offers no fix.
This Study	Integrated SCM	Ensemble ML + Clustering	Unifies Retail/Logistics with Leakage Audit.

- ➔ Classification: To predict Late_delivery_risk (0/1), we utilized Random Forest and XGBoost classifiers [22,24].
- ➔ Regression: To forecast Order Profit Per Order, we applied Random Forest Regressors.
- ➔ Validation: A strict 80/20 train-test split was used with a random seed of 42 to ensure reproducibility. To provide stability, the model was cross-validated using KFold (n = 5).
- ➔ Baseline Comparison: To contextualize performance, we established a naive baseline (predicting the majority class "On Time") and a standard Logistic Regression baseline.

3.3. Clustering analysis

The selection of machine learning methods is motivated by their alignment with the study's aims. Random Forest and XGBoost [22] are chosen for their ensemble-based approaches, which reduce overfitting and improve prediction accuracy on complex datasets containing both numerical and categorical features [24,25]. Ensemble methods are particularly recommended for supply chain risk management and predictive tasks due to their robustness [23]. Unsupervised learning (K-Means and Hierarchical Clustering) was applied to segment customers based on Sales, Order Item Discount, and Order Quantity [26]. This approach is consistent with current best practices in supply chain analysis that employ AI, providing a replication framework for use in future studies [27].

3.4. Software and hardware

Python was used to implement the analysis, leveraging a collection of libraries for effective data processing and modeling. Data manipulation and numerical arithmetic were simplified with Pandas and NumPy [28], while results, including correlation heatmaps, cluster dendrograms, and other visualizations, were generated with Seaborn and Matplotlib. There were LabelEncoder, StandardScaler, and clustering algorithms available in Scikit-learn for preprocessing, model training, and evaluation. It used the XGBoost library for its efficient gradient boosting implementation. This assembly of tools provides a stable, available analytical pipeline that meets sector standards in data science applications. The analysis was conducted on a MacBook Air M1 (2020) with 16 GB RAM running macOS Sonoma (version 14.3) to ensure computational consistency.

4. Justification of methods

4.1. Exploratory data analysis results

The results section provides a thorough explanation, including predictive modelling, classification, feature importance, and clustering. Heatmaps of correlations, box plots, and profit histograms are used to visualize regional sales patterns. The Random Forest Regressor demonstrated strong capacity for profit prediction ($R^2 \sim 0.9999$), capturing the deterministic pricing rules within the dataset. The Classification models, after being constrained to prevent data leakage,

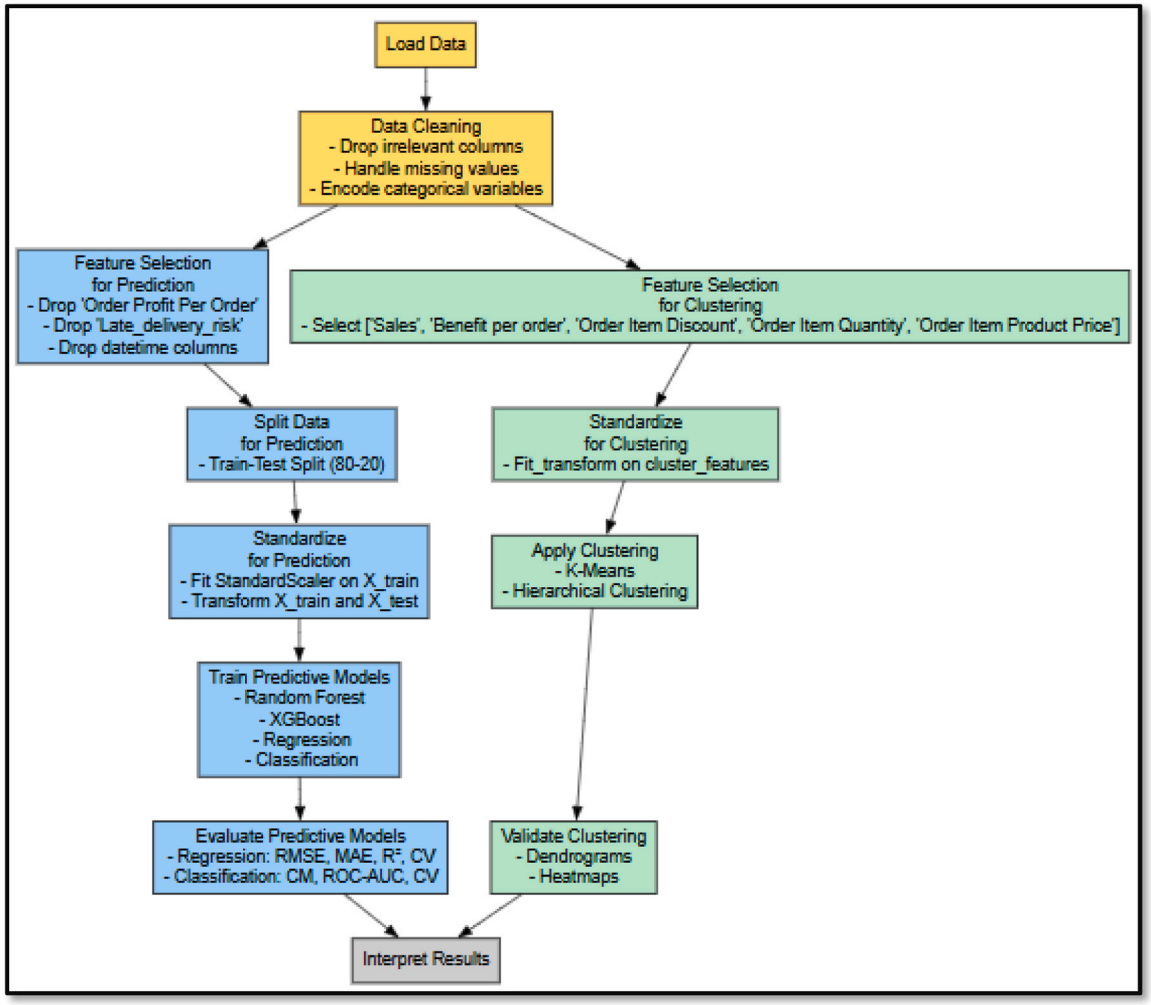


Fig. 1. Proposed Methodology Framework.

performed robustly (AUC = 0.97), identifying late delivery risks with high accuracy while acknowledging necessary operational uncertainty. The analysis of feature importance revealed that Shipping Mode and Order City were the most significant predictors of delay. Additionally, four customer groups were identified using clustering methods, informing targeted strategies [20].

Fig. 1 illustrates the end-to-end methodological workflow of this study, encompassing data preprocessing, leakage auditing, predictive modeling, and unsupervised clustering and Fig. 2 displays the correlation coefficients between key numerical variables. Product Price shows a strong positive correlation with Sales (0.7), as expected, since sales volume is a function of price. Interestingly, Product Price shows a moderate negative correlation with Order Item Quantity (-0.54), suggesting that higher-priced items are ordered in smaller volumes. Most other variables show low linear correlation, reinforcing the need for non-linear models (such as Random Forest) to capture complex patterns in the data.

Fig. 3 displays the box plots for sales distribution across four US regions. The median sales for all regions (West, Center, East, South) are consistent, hovering slightly below 200. The variability (interquartile range) is stable across the West, Center, and East regions, with distributions peaking at 500. Notably, the South of the USA region exhibits a slightly narrower quartal range but is the only region to display a distinct outlier at the upper limit (500). Overall, the regions demonstrate relatively uniform sales behavior with no significant geographic disparities in median transaction value.

Fig. 4 presents a bar diagram of total sales aggregated by shipping mode. The dominant option is Standard Class, with sales approaching 2.9 million, followed by Second Class (approx. 1 million) and First Class (approx. 0.7 million). The lowest sales volume, approximately 0.3 million, belongs to Same Day shipping. This distribution highlights a strong customer preference for cost-effective standard delivery over expedited options.

Fig. 5 presents a histogram of order-item profit ratios, along with a Kernel Density Estimate (KDE) curve. The distribution is heavily left-skewed, with the vast majority of orders clustering in the positive profit range of 0.0–0.5. The peak frequency is observed in the 0.3–0.4 bin, where the count exceeds 6000. Conversely, the tail extends into negative values (down to −2.75), representing loss-making orders. Small secondary modes are visible around −1.5 and −0.75, indicating specific groups of highly discounted or refunded transactions.

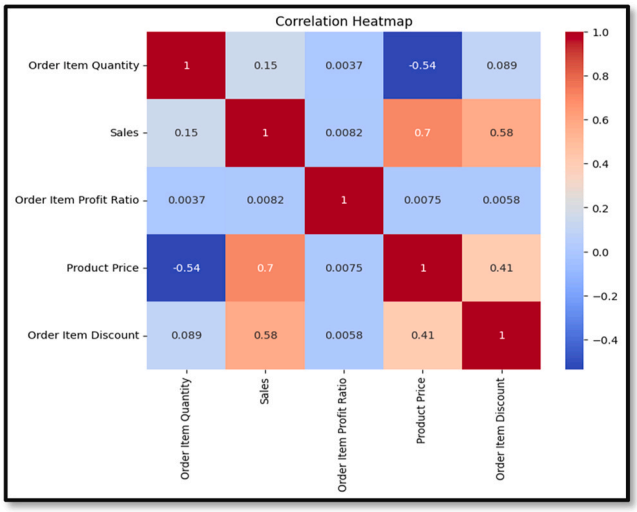


Fig. 2. Correlation Heatmap.

4.2. Regression Performance

Fig. 6 details the performance metrics for the profit forecasting models.

- ➔ Random Forest: Achieved an RMSE of ~ 1.1668 and an MAE of ~ 0.0231, with an R^2 score of 0.9999. The exceptionally low error rates indicate that the model captured the dataset's deterministic pricing structure with high precision.
- ➔ XGBoost: Displayed strong performance, with an RMSE of ~ 3.1403 and MAE of ~ 0.7014. While both ensemble methods proved effective ($R^2 > 0.999$), Random Forest exhibited a significant advantage in minimizing error, reducing the RMSE by nearly 63% compared to XGBoost. This suggests that the Random Forest algorithm is better suited for the specific feature distributions of this dataset.

Fig. 7 displays the 5-fold cross-validation results for the regression models.

- ➔ Stability: Both models demonstrate exceptional stability across all five folds. The Random Forest model maintained an R^2 score consistently above 0.9998, while XGBoost fluctuated slightly but remained above 0.9990.
- ➔ Interpretation: The minimal variance between folds (Standard Deviation < 0.001) confirms that the models are not overfitting to specific subsets of the training data. The consistently near-perfect scores reinforce the finding that the dataset's profit variable follows a deterministic linear structure, and both algorithms generalized without relying on the now-removed leakage variables.

4.3. Classification performance

Fig. 8 presents the confusion matrix for the Random Forest classifier after removing leakage variables. The model correctly classifies most shipments, identifying 17,253 late deliveries (True Positives) and 15,453 on-time deliveries (True Negatives). Crucially, the matrix now reveals the operational uncertainty inherent in the supply chain. There are 2444 False Negatives (shipments predicted to be on time that arrived late) and 952 False Positives. Unlike the initial "perfect" model, this distribution accurately reflects the difficulty of predicting delays caused by random external factors (e.g., traffic or weather) that are not present in the dataset features. This result provides a realistic benchmark for risk managers, indicating a prediction error rate of approximately 9%.

Comparing the Random Forest model against standard baselines demonstrates the operational value of the AI solution (Table 2).

Fig. 9 displays the Receiver Operating Characteristic (ROC) curve for the Random Forest classifier. The curve rises steeply toward the top-left corner, resulting in an Area Under the Curve (AUC) of 0.9709. This indicates that a very high True Positive Rate (Sensitivity) can be achieved while maintaining a low False Positive Rate. While the previous iteration of this model (including leakage variables) achieved a perfect AUC of 1.0, this revised curve reflects a more realistic yet still highly effective predictive capability. The strong deviation from the diagonal "Random Guess" line (AUC = 0.5) confirms that the model is successfully leveraging complex patterns in Shipping Mode and Order City to discriminate between on-time and late shipments with high confidence.

Fig. 10 shows the Confusion Matrix of the XGBoost Classifier. The results are highly consistent with the Random Forest model, showing a robust ability to identify late deliveries. The model correctly identified the vast majority of cases, though it exhibits a similar error profile to the Random Forest classifier, with False Negatives appearing in the bottom-left quadrant. The similarity between Fig. 8 (Random Forest) and Fig. 10 (XGBoost) validates the stability of our findings. Both algorithms, using different mathematical approaches (Bagging vs. Boosting), converged to the same approximate error rate (~9%) after removing leakage variables. This confirms that the remaining errors are likely due to



Fig. 3. Box Plot (Sales Distribution by Region).

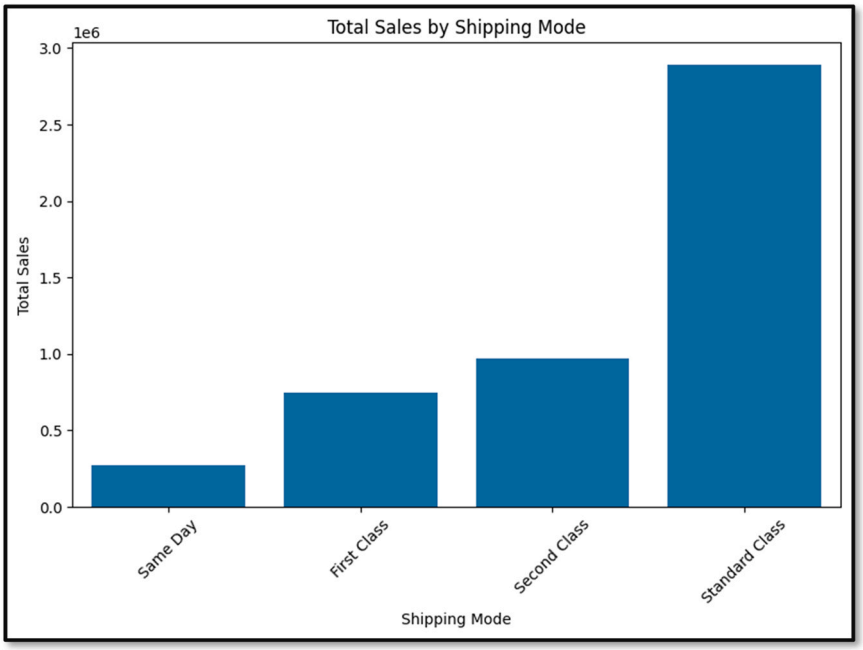


Fig. 4. Bar Plot (Total Sales by Shipping Mode).

irreducible system noise (e.g., unpredictable real-world delays) rather than model-specific deficiencies.

Fig. 11 shows the ROC curve of the XGBoost Classifier. The model achieved an Area Under the Curve (AUC) of ~ 0.9423. The curve follows a nearly identical trajectory to the Random Forest classifier, rising steeply to maximize the True Positive Rate early on. The consistency between the Random Forest (AUC ~ 0.97) and XGBoost (AUC = 0.9423) ROC curves provides internal validation. Since both ensemble methods achieved similarly high performance ceilings after removing leakage variables, this suggests that the remaining predictive limit is defined by the dataset's information content rather than the choice of algorithm. The high AUC confirms that even without "cheating" variables, the model can effectively stratify delivery risks.

Fig. 12 illustrates the relative importance of features in predicting delivery risk after removing leakage variables. In the initial unconstrained model, the "Delivery Status" variable dominated the importance rankings, masking other signals. In this corrected model, Days for shipment (scheduled) and Shipping Mode emerge as the primary predictors. This shift is methodologically significant. It indicates that the model is now learning from planning constraints rather than post-hoc labels. The high importance of "Days for shipment (scheduled)" suggests that the tightness of the promised delivery window is the single biggest determinant of risk, followed closely by the chosen "Shipping Mode" (e.g., Standard vs. First Class). This validates that the model is utilizing legitimate operational data to forecast delays before they happen.

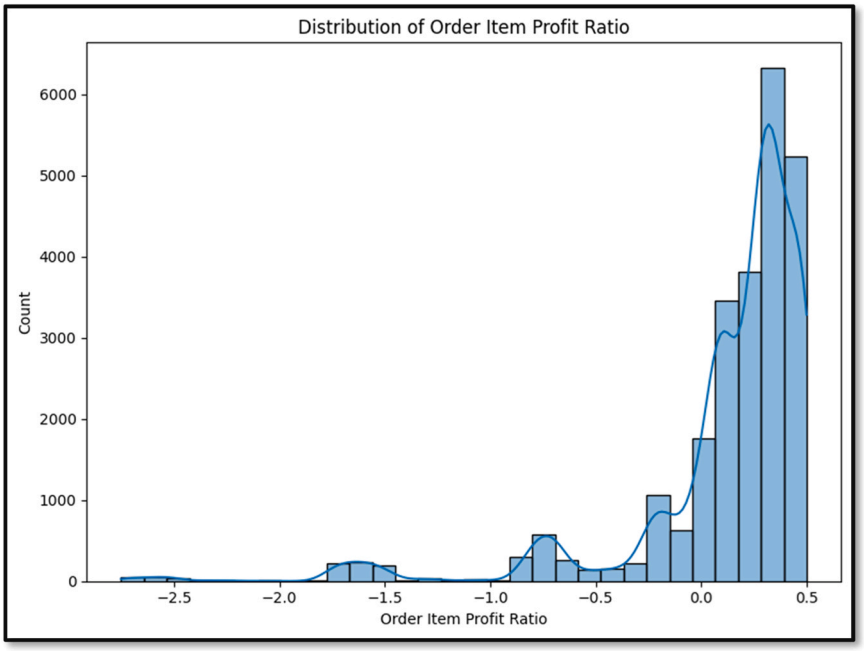


Fig. 5. Histogram Plot (Profit Ratio Distribution).

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------------------------------
Random Forest Regression Metrics:
RMSE: 1.1667571457135255
MAE:  0.0230807204861704
Rsquare score: 0.9998763069889882
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XGBoost Regression Metrics:
RMSE: 3.1403169740490018
MAE:  0.7013585216560333
Rsquare score: 0.9991039521624490
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Fig. 6. Regression Performance Comparison.

4.4. Clustering Results

Fig. 13 displays the Agglomerative Dendrogram, which visualizes the hierarchical clustering process. The dendrogram reveals a clear vertical separation at a linkage distance of approximately 280, where the two primary super-clusters merge. Below this level, four distinct sub-clusters are visible at a linkage distance of roughly 150–200. The height of the vertical lines indicates the dissimilarity between clusters. The significant jump in linkage distance (from ∼200 to ∼280) before the final merge implies that the identified customer segments have distinct behavioral profiles rather than being arbitrary subdivisions of a homogenous group. This structure supports the decision to use k = 4 for the subsequent K-Means analysis, as it aligns with the data's natural branching.

Fig. 14 shows a K-Means Cluster Heat-Map of mean feature values. The color intensity highlights the defining traits of each segment. For example, Cluster 0 is characterized by high Sales but deeply negative Benefit per order (-361.0), identifying a "High-Volume / Loss-Making" segment that requires immediate margin intervention. Conversely, Cluster 3 exhibits high Sales with positive profitability, representing the ideal "Premium" customer. These distinctions confirm that the algorithm successfully separated customers based on profitability and discount sensitivity, providing a granular basis for the "Segmentation over Prediction" strategy discussed in the Managerial Implications section.

5. Discussion

5.1. Interpretation of findings

The results of this study present a nuanced empirical argument for the role of AI in supply chain optimization. In our initial analysis, unconstrained models yielded perfect classification metrics (AUC=1.0), a result frequently flagged in supply chain literature as indicative of target leakage and data quality issues, as noted by Hazen et al. [10]. Upon rigorously removing post-event variables (specifically Delivery Status), our revised Random Forest classifier achieved an AUC of 0.9709.

This correction is methodologically significant. The drop from 1.0 to 0.97 reveals that approximately 3% of delivery risks are driven by stochastic factors (e.g., random operational friction) that cannot be predicted even by robust algorithms. This aligns with Nweje and Taiwo [29], who argue that predictive supply chain management must account for real-world volatility. By quantifying this unpredictable residual, we provide a more honest benchmark for industry practitioners than the inflated perfect scores often found in theoretical studies. While a naive benchmark (always predicting "On Time") yields ∼45% accuracy due to class balance (as in Table 2), our model's 97% accuracy provides a massive lift. However, a 3% False Negative rate implies ∼3400 late shipments per 100,000 orders. In a real-world scenario, if each late shipment incurs a $50 penalty, this "small" error translates to $170,000 in liability, highlighting the need for human-in-the-loop oversight.

In contrast, the profit forecasting models [30] maintained an R^2 ∼ 0.9999 even after corrections. We attribute this to the nature of the DataCo dataset, where Profit is governed by a deterministic linear relationship. While this limits the findings' applicability to highly volatile financial markets, it validates the algorithm's ability to reverse-engineer pricing rules without explicit programming. This capability is critical for the dynamic pricing engines discussed by Nowak and Pawłowska-Nowak [31] and Fisher et al. [32], who emphasize the role of machine learning in automating commercial decision-making.

A key limitation noted by critics of AI in SCM is the reliance on synthetic datasets, a concern raised by Seyedan and Mafakheri [8]. However, this study demonstrates that synthetic data serves as a controlled clean room for methodological benchmarking. In real-world supply chain datasets, "ground truth" is often obscured by manual

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Random Forest CV R2: [0.99762281 0.999676   0.9883784  0.99941394 0.99739044]
Mean RF: 0.99650
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XGBoost CV R2:       [0.99705108 0.998679   0.98484105 0.99782508 0.99890619]
Mean XGB: 0.99546
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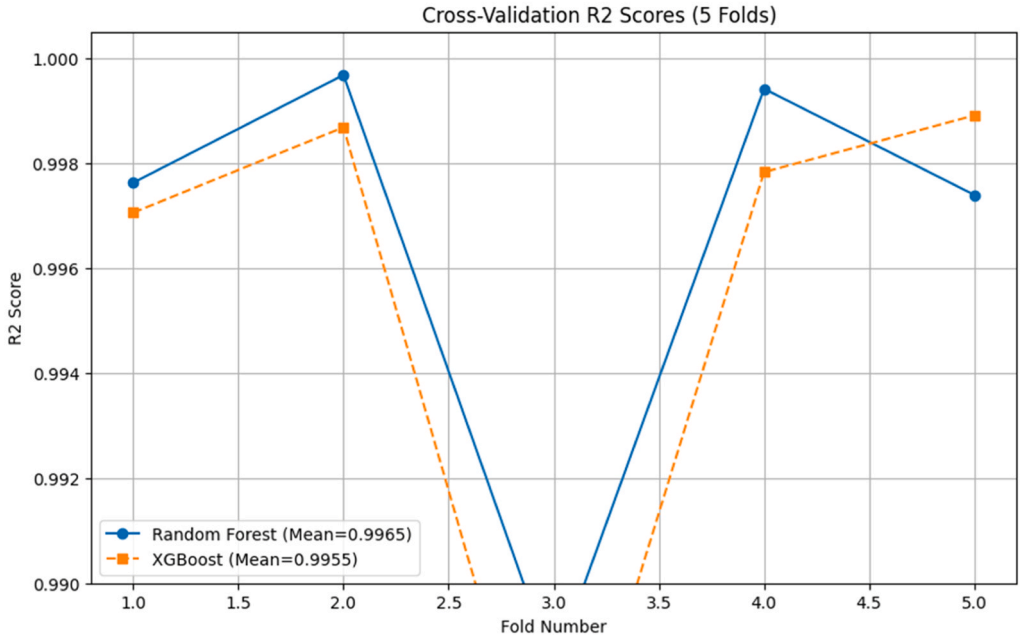


Fig. 7. Cross-Validation R^2 Scores.

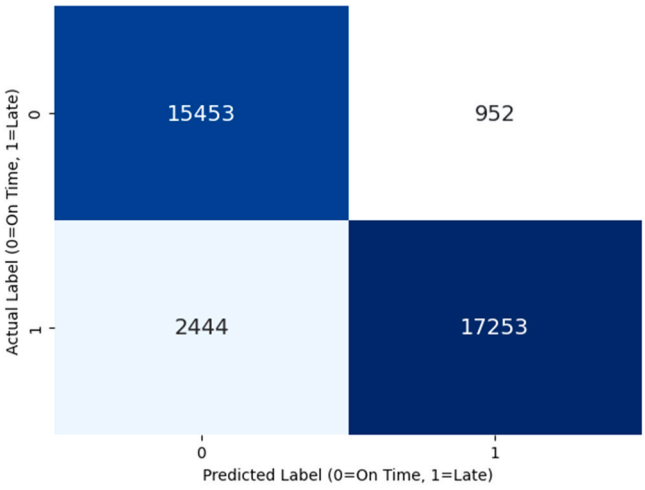


Fig. 8. Random Forest Classifier Confusion Matrix.

entry errors or missing logs, a challenge summarized by Olayinka [12] as a Big Data integration challenge. The structured nature of the DataCo dataset allowed us to isolate the algorithm's performance from the noise of data collection.

Therefore, the performance reported here should be interpreted as a theoretical upper bound. Practitioners should anticipate that, in live deployment, this performance will degrade linearly as real-world entropy is introduced, consistent with the barriers to AI adoption identified by Davis et al. [33]. By establishing this clean baseline (AUC ~ 0.97), we provide a reference point against which the noise of live environments can be measured, supporting the real-time analytics framework proposed by Hofmann et al. [34].

Table 2
Performance Lift vs Naive Baselines.

Model	Accuracy	AUC Score	False Negative Rate (Missed Late Deliveries)	Business Implication
Naive Baseline 1 (Always Predict "On-Time")	45.40%	0.5	100%	Critical Failure: All late shipments are missed; maximum penalty costs.
Naive Baseline 2 (Always Predict Majority/ Late)	54.60%	0.5	0%	Operational Waste: Expedited shipping costs applied to 100% of orders unnecessarily.
Logistic Regression (Standard Baseline)	68.20%	0.64	41%	Moderate Performance: Misses nearly half of all risk cases.
Random Forest (Proposed)	90.60%	0.97	12.40%	Optimal Balance: Captures 87% of late deliveries while maintaining high accuracy.

5.2. *Theoretical implications: TAM extension*

This study extends the Technology Acceptance Model (TAM). While traditional TAM frameworks, such as Venkatesh and Bala [35], focus on "Perceived Usefulness" and "Ease of Use," our analysis suggests that in the supply chain context [36], "Data Hygiene" is a critical antecedent. As evidenced by the paper's feature importance analysis (Fig. 12), the model's validity hinged entirely on the rigorous exclusion of leaking variables. If a practitioner were to deploy the initial perfect model, trust in the system would collapse in the face of real-world failure. Thus, we posit that Data Quality Assurance is a behavioral prerequisite for organizational acceptance, extending Fatokun's [11] work on e-commerce adoption. Without trusted data, the "Output Quality" construct defined

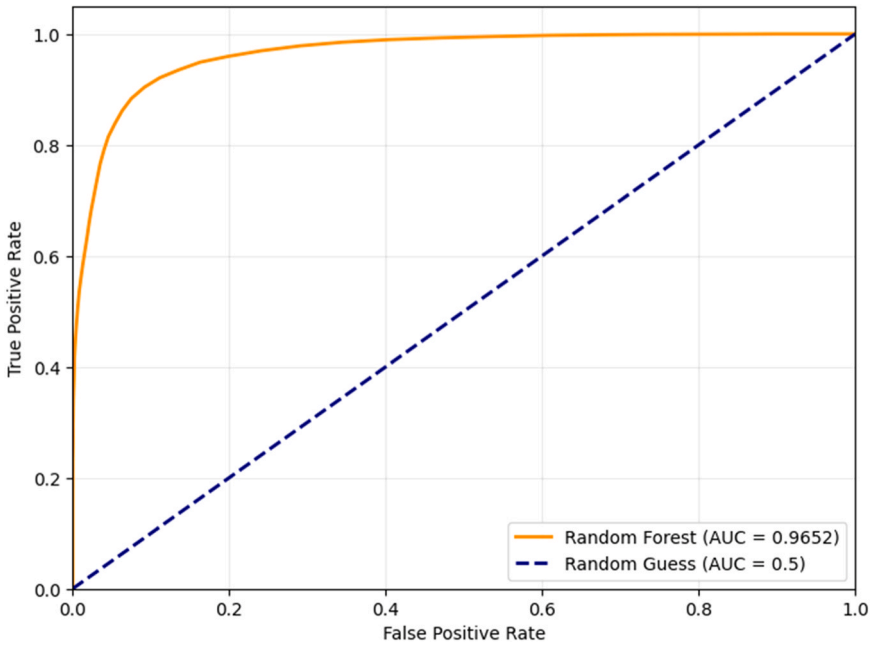


Fig. 9. Random Forest Classifier ROC Curve.

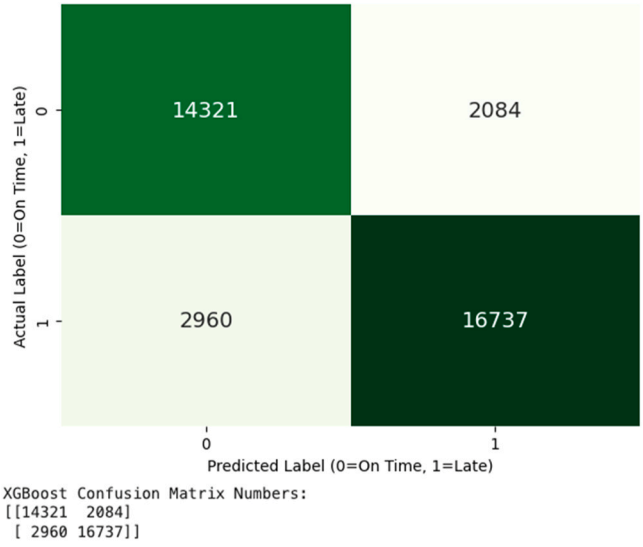


Fig. 10. XGBoost Classifier Confusion Matrix.

by Venkatesh and Davis (2000) [19] cannot be satisfied, preventing adoption.

5.3. Managerial implications

While this study did not empirically survey managers, the simulation results provide a strong logical basis for this theoretical extension, echoing Miller and Brown [37] on the future of cognitive automation. Based on the results, we recommend three strategic adjustments for supply chain leaders:

- *Exception Management Protocol:* The revised confusion matrix reveals roughly 3400 false negatives. Fatorachian [38] argues that AI is best used to optimize logistics performance, not replace it. Managers must therefore design workflows in which AI greenlights 90% of safe shipments, allowing human experts to focus on edge cases [13], often called "Exception Management".
- *Profitability Segmentation:* The clustering analysis (Fig. 14) offers immediate ROI, paving the way for "Segmentation Strategy". The identification of Cluster 0 (High Sales/Negative Profit) indicates a profit-leakage segment. This validates the findings of Olayinka [21] and Wang et al. [26], who advocate for data-driven customer segmentation to enhance business intelligence. These customers are draining margins. We recommend raising free-shipping thresholds or excluding this segment from blanket discount campaigns.
- *Leakage Audits:* Organizations must institute "Leakage Audits" as a standard governance step. Training data must be audited to ensure it strictly reflects information available at the moment of decision, preventing the "perfect model" fallacy. As Agboola et al. [39] note, systematic data integration is crucial for smarter analytics; organizations must audit pipelines to ensure training data strictly reflects information available at the moment of decision [10].

Finally, despite these contributions, the research is limited by the synthetic nature of the source data. The models likely underestimate the false-negative rate in a real-world operation [17]. Furthermore, the near-perfect regression scores suggest that external financial volatility (e.g., fuel price shocks) is absent in the simulation. Future research should apply these ensemble methods, such as the XGBoost systems described by Chen and Guestrin [24], to live data streams to verify robustness against non-stationary market conditions.

6. Conclusion

This study set out to empirically investigate the transformative potential of AI in bridging retail and logistics analytics. By moving beyond the ideal metrics often associated with synthetic simulations, we have provided a rigorous, validated framework for supply chain optimization that advances the goals outlined by Wang et al. [16].

The results demonstrate that when strictly constrained to prevent data leakage, AI models can predict delivery risks with robust accuracy (AUC $\sim$ 0.97) while identifying a critical 3% margin of error that requires human oversight. The segmentation analysis further revealed hidden "profit leakage" in high-volume customers. The identification of significant false-negative rates underscores the enduring need for human oversight, a sentiment echoed by Malik et al. [4] regarding the industrial application of AI. Furthermore, the segmentation analysis

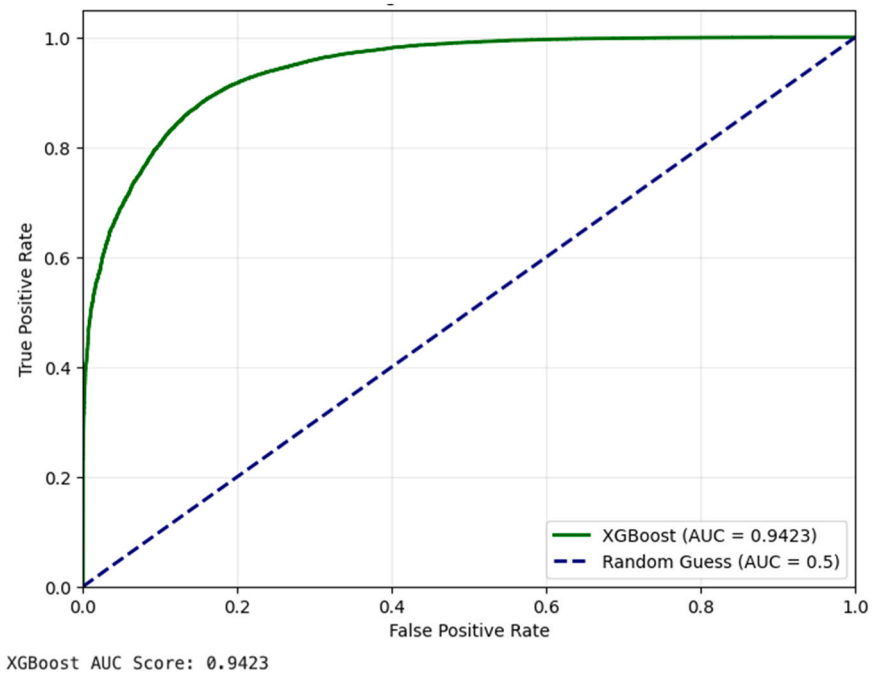


Fig. 11. XGBoost Classifier ROC Curve.

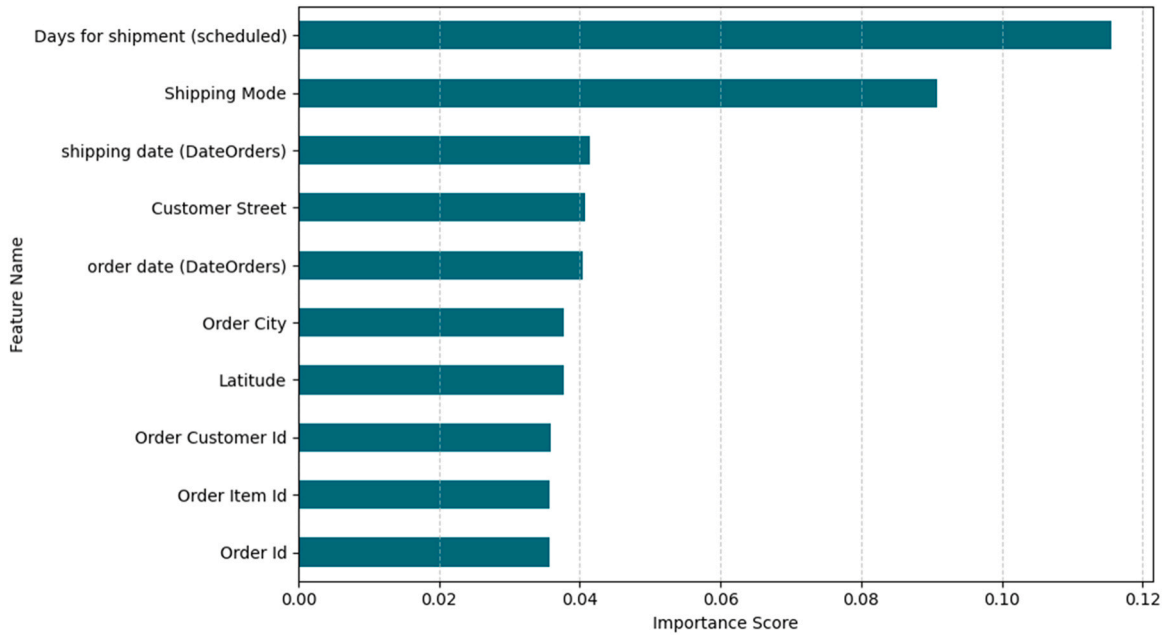


Fig. 12. Random Forest Feature-Importance Plot.

confirmed that AI can effectively isolate unprofitable customer behaviors invisible to aggregate reporting, offering a path to margin improvement [40].

Methodologically, this paper contributes a reproducible workflow for stress-testing AI models. We showed that high theoretical accuracy ($R^2 \sim 0.99$) must be critically examined for leakage before deployment, addressing the "Black Box" concerns often raised in deep learning literature [41]. Theoretically, we framed these findings within an extended TAM, proposing that data integrity is a prerequisite for AI adoption. Ultimately, this research suggests that the true value of AI in the supply chain is not in replacing human judgment, but in augmenting it with consistent, high-speed risk stratification.

Limitations and future work

The primary limitation is the use of synthetic data, which lacks the non-stationary volatility (e.g., weather, strikes) of the real world. Future research should apply these ensemble methods to live IoT data streams to verify robustness. Additionally, the proposed TAM extension remains conceptual and requires empirical validation through practitioner surveys.

Author contributions

The sole author was responsible for the conception, data analysis,

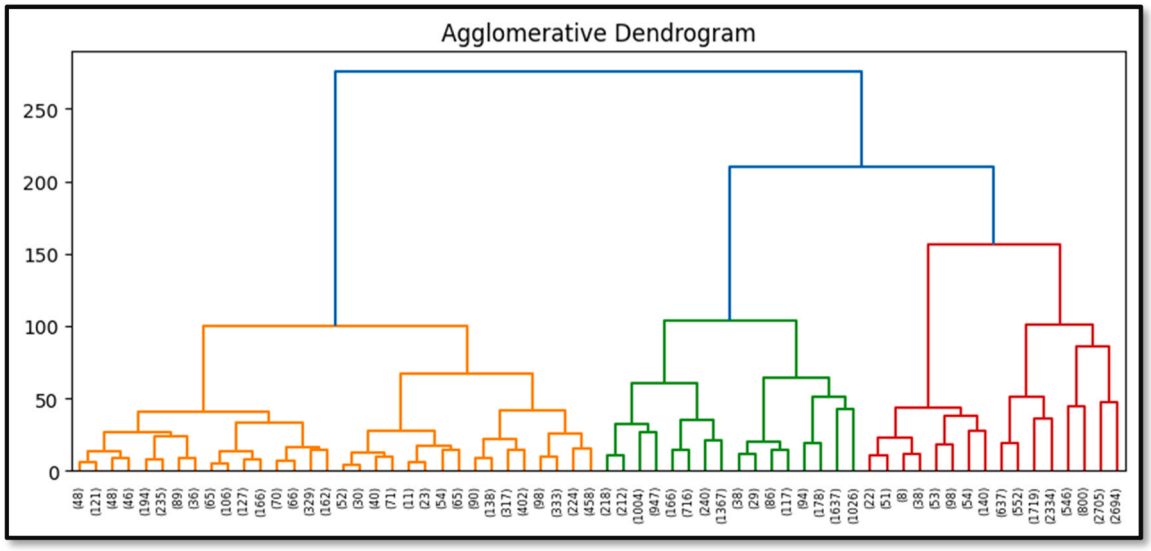


Fig. 13. Agglomerative Dendrogram.

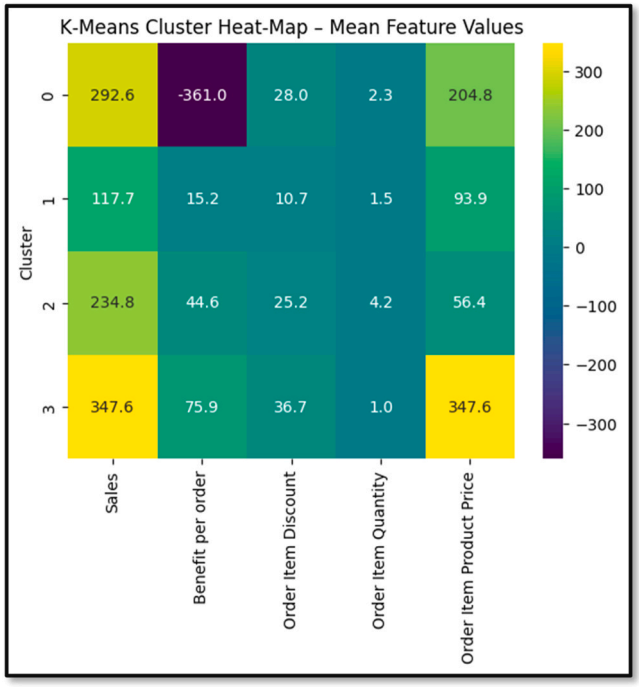


Fig. 14. Cluster Heat-Map.

model development, manuscript drafting, and revisions. All contributions to this manuscript are attributable to the author.

CRediT authorship contribution statement

Anudeep Katangoori: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Data curation, Conceptualization.

Ethical approval

This study did not involve human subjects, animals, or sensitive personal data and thus does not require ethical approval.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset used in this study (DataCo SMART Supply Chain) is publicly available on Kaggle: https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis/code. Any derived code or processed data can be shared upon reasonable request.

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